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DESIGN OF A HEALTH, SAFETY AND EQUIPMENT CENTRALISED MONITORING APPLICATION FOR KHWEZI MINE

Group 8

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1 Abstract

This report presents the development of a Mining Health, Safety and Equipment Monitoring Application for Khwezi Mining using Python, Streamlit and Pandas. The application was designed to centralize worker safety information, incident records, equipment condition data, and maintenance information in an interactive platform. Its main features include role-based access control, worker safety monitoring, incident management, equipment condition monitoring, risk assessment and automated alerts. Testing was conducted using synthetic data to evaluate the application's functionality and monitoring capabilities. The results identified 86 cases of personal protective equipment (PPE) non-compliance among 400 workers, with an average PPE compliance rate of 90.4 percent. Of the 60 equipment units assessed, 44 were classified as normal, 11 as warning and five as critical. Average equipment availability was 90.2 percent . Furthermore, 58 of the 160 valid risk assessments were classified as high or critical. These findings demonstrate the application's ability to consolidate operational information, identify potential risks and support decision-making. However, the application is an educational prototype, and its thresholds and results should not be interpreted as official mine safety standards or manufacturer limits.

2 Introduction

Mining is a major contributor to South Africa's economy, supporting employment, industrial development and economic activity [1]. It also exposes workers to significant occupational health and safety risks and relies on large, complex mobile and fixed equipment [2]. Managing these risks requires timely, accurate information on safety incidents, unsafe conditions, worker observations, equipment condition and maintenance status.

At Khwezi Mining, this information is collected from separate sources: safety observations, incident reports, equipment inspections, maintenance records, alerts and notification reports and operator reports. As a result, supervisors and engineers cannot easily get a consolidated view of the operation's health and safety status of the condition of its critical equipment. Management therefore requested a centralised prototype application that collects and analyses this data, identifies safety risks, tracks equipment condition, flags equipment needing maintenance and generates alerts when predefined conditions are exceeded.

This report documents the design, development and testing of the *Mining Health, Safety and Equipment Monitoring Application*, built in Python by a group of junior mining engineers. The application brings together role-based access control, worker safety monitoring, a safety incident database, equipment condition and maintenance monitoring, risk assessment, automated alerts, a dashboard and interactive data analysis in a single user-friendly platform. The thresholds used are educational prototype values only and do not represent

manufacturer or mine safety limits.

3 Background

Digital technologies, sensors and data analytics now allow mining companies to monitor health, safety and equipment conditions far more effectively than manual inspections and paper-based records [3]. Computer applications can collect, process, visualise and interpret operational data, helping supervisors identify abnormal conditions, prioritise maintenance and recognise safety risks before they escalate [3, 2]. These developments form part of the wider modernisation of the South African mining sector [1].

Two engineering concepts underpin the application. The first is **risk assessment**, where a risk score is calculated as:

$$RiskScore = Likelihood \times Consequence,$$

with both factors rated from 1 to 5 and the result classified as Low (1–4), Medium (5–9), High (10–16) or Critical (17–25). The second is **equipment availability**, calculated as:

$$Availability = \frac{OperatingTime}{OperatingTime + Downtime} \times 100,$$

which is used to identify equipment with excessive downtime or low availability. Condition monitoring compares parameters such as temperature and vibration against Normal, Warning and Critical thresholds to trigger alerts.

The application was developed with Python, using Streamlit for the interactive interface, Pandas for data handling and analysis, and SQLite for data storage. Development followed the full workflow from problem to solution: flowchart and pseudocode for each function, followed by Python implementation, data analysis, visualisation and a working application.

4 Methodology

The methodology started with identifying the problems faced by Khwezi Mine in managing worker safety, incidents, equipment conditions and maintenance records seperately. The requirements were then identified to develop an application that could bring this information together and make it easier to monitor safety and equipment performance. Python, Streamlit, and Pandas were used to develop the application, process the data, and display the results on an interactive dashboard. Different algorithms were used to calculate risk scores based on likelihood and consequence, monitor equipment temperature and vibration, and assess worker PPE compliance and equipment availability. An alert system was also developed to identify critical and warning conditions that may require

attention. The application includes role-based access control, allowing administrators, safety officers, mining engineers, maintenance engineers, and managers to access different sections according to their roles. Finally, the main features, calculations, data displays, and login functions were tested to check that the application worked as expected. The thresholds used in the application are for prototype purposes only and should not be treated as official mine safety standards or equipment manufacturer limits. In addition, the data used to test the app was synthetic data generated.

5 Results and Analysis

5.1 Overall Application Performance

The health, safety and equipment monitoring application was developed using Python and Streamlit to demonstrate the integration of various monitoring functions into one platform. The application incorporates worker health and safety monitoring, incident management, equipment condition monitoring, maintenance management, risk assessment, and automated alerts based on user.

5.2 Access Control

The access control functions implemented user authentication and role-based permissions across five user caterogies (Administrator, Safety Officer,Mining Engineer, Maintenance Engineer and the Manager). Following authentication, the application displays navigation options according to the assigned role, restricting access to functions outside the user's responsibilities. For example, maintenance were allocated equipment monitoring functions, while administrators were granted additional user-management permissions. This access control feature of the application improves information security, accountability and separation of responsibilities in the centrilised Khwezi Mine monitoring system.

5.3 Analysis of the Worker Safety Results

Worker safety monitoring identified 86 PPE non-compliant cases out of 400 workers. The average PPE compliance percentage was 90.4. Additionally, 21 Workers were classified as high or critical risk. There were a total of 380 safety incidents, 69 of which were highly/critically severe, and only 65 incidents were open or under investigation. The total lost time injuries were 34. Tehse findings indicate the application's ability to identify safety concerns and where the focus the necessary corrective/preventative measures to improve the safety of workers.

5.4 Equipment Monitoring and Risk Assessment

The data set contained a total of 60 equipment units. Equipment condition monitoring functions show 44 machines complying with what was defined as

normal equipment conditions, 11 with a condition warning, and five in a critical condition that requires immediate inspection according to predefined temperature and vibration thresholds. The average equipment availability is 90.2 percent, although seven machines are overdue for maintenance, nine are due for maintenance and five are under maintenance. Notably, TRK-009 recorded 74 percent availability with 561.4 hours of downtime, demonstrating how individual equipment may differ significantly from fleet averages. The results from the dataset show that 58 of the 160 valid assessments are classified as high or critical, indicating conditions that require prioritised investigation.

6 Conclusion

The Mining Health, Safety and Equipment Monitoring Application was successfully developed as a prototype to integrate worker safety, incident management, equipment condition monitoring, maintenance management and risk assessment into a centralised platform. The application uses Python, Streamlit and Pandas to process and display information, while role-based access control provides different levels of access according to user responsibilities.

Testing using synthetic data demonstrated the application's ability to identify PPE non-compliance, classify worker and equipment risks, monitor equipment availability and highlight maintenance requirements. The results identified 86 PPE non-compliant cases, five equipment units in critical condition and 58 high- or critical-risk assessments among 160 valid assessments. These findings demonstrate how the application can help users identify areas requiring further investigation and corrective action.

Overall, the prototype demonstrates the potential of digital monitoring systems to improve the organisation and accessibility of mining health, safety and equipment information. However, its effectiveness in an actual mining environment would depend on reliable operational data, appropriate security measures and the validation of its calculations and alert thresholds. Further development and testing would therefore be required before the application could be implemented in an operational mine.

7 Recommendations

The following improvements are recommended:

1. Data integration:Connect the application to actual mine safety and equipment records for more accurate monitoring.

2. Threshold validation: Ensure that the risk assessment methods and equipment limits are reviewed by qualified professionals.

3. Security Strengthen authentication and access controls to protect sensitive information.

4. Automated alerts: Improve notifications for critical equipment conditions, safety risks, and overdue maintenance.

5. Further testing:Test the application with real operational data to evaluate its precision and reliability.

These improvements could enhance the application's effectiveness and support its future development for mining operations.

References

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